# Practice workbook · Substitute into a given equation, rearranging and converting units

A physical-style relationship such as speed, distance and time, energy from heating water or force times distance, cardiac output or reaction time is given or implied.

9 question parts: 1 worked-example retry and 8 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-SUBSTITUTE-FORMULA

# 1. Question 3 · part b.iv · 3 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.iv only.

**3** (a) Describe what is meant by a specialised cell.

**(2)**

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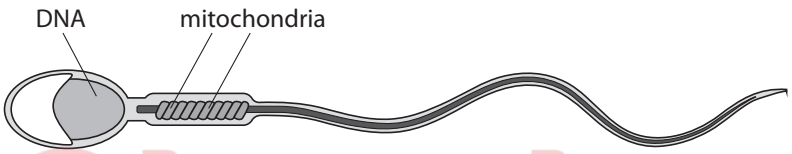
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(b) The sperm cell is an example of a specialised cell.

The diagram shows a sperm cell.

NOT TO SCALE




(i) The sperm cell in the diagram has an actual length of 0.0060 mm.

The length of the image of the sperm cell is 84 mm.

Calculate the magnification of the sperm cell.

**(2)**

magnification = × .....

(ii) The head of the sperm contains DNA.

Give the difference in the amount of DNA found in the head of the sperm and in a nucleus of a body cell.

**(1)**

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# 1. Question 3 · part b.iv · 3 marks

June 2023 Paper 2R

Worked-example retry · Attempt part b.iv only.

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(iii) Explain why the sperm cell has many mitochondria.

**(2)**

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(iv) A sperm cell travels at a speed of 0.070 mm/s.

It travels a distance of 180 mm to fertilise an egg.

Use this equation to calculate the time it takes for the sperm to reach the egg.

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Give your answer in minutes.

**(3)**

time taken = ..... minutes

(c) An electron microscope can be used to view structures in specialised cells.

Explain why an electron microscope is more useful than a light microscope to view cell structure.

**(2)**

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**(Total for Question 3 = 12 marks)**

## 1. Question 3 · part b.iv · 3 marks · Working and answer

Working and answer

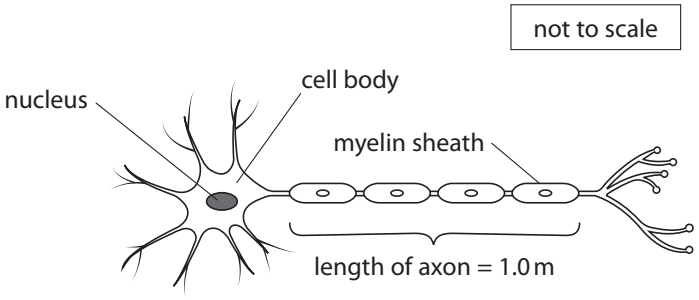
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## 2. Question 5 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

**5** The diagram shows a motor neurone.




(a) (i) The actual diameter of the nucleus of the motor neurone is 12 µm.

A student observes the motor neurone using a ×50 microscope lens.

Calculate the diameter of the image seen after magnification.

Give your answer in millimetres.

**(2)**

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diameter = ..... mm

(ii) Neurones transmit electrical impulses from one part of the body to another.

Suggest one function of the myelin sheath surrounding the axon.

**(1)**

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(b) Nerve impulses travel at an average speed of $1.2 \times 10^8$ µm per second.

Hormones travel at an average speed of $1.5 \times 10^6$ µm per second.

(i) Explain this difference in speed.

**(2)**

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## 2. Question 5 · part b.ii · 3 marks

January 2020 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Calculate the time taken for a nerve impulse to travel along the axon of the motor neurone.

**(3)**

time taken = ..... s

**(Total for Question 5 = 8 marks)**

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Working and answer

# 3. Question 3 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

**3** (a) The nervous system is adapted to transfer information rapidly from one part of the body to another.

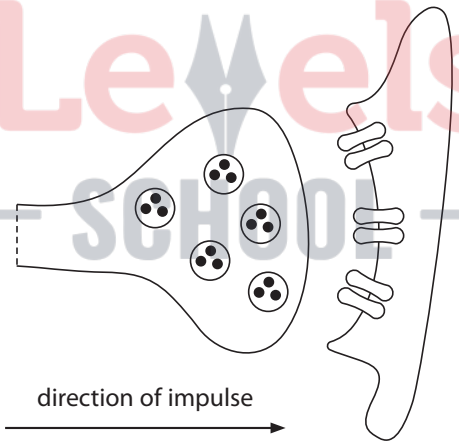
Complete the passage about the nervous system by giving the correct words.

**(4)**

A change in the external environment is detected by ..... in sense organs. These structures convert stimuli into electrical impulses which then travel along ..... neurones to the central nervous system.

The central nervous system coordinates the incoming information using ..... neurones. These neurones then transfer the electrical impulses to ..... neurones.

(b) The diagram shows the junction between two neurones.




(i) State the name of the junction between two neurones.

**(1)**

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## 3. Question 3 · part c · 2 marks

June 2022 Paper 2

Further practice · Attempt part c only.

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(ii) Describe how an impulse reaching the end of one neurone produces an impulse in the next neurone.

**(2)**

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(c) A nerve impulse travels at a speed of 120 metres per second.

Calculate the distance that this nerve impulse will travel in 5.0 minutes.

**(2)**

distance = ..... m

**(Total for Question 3 = 9 marks)**

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## 3. Question 3 · part c · 2 marks · Working and answer

Working and answer

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## 4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

**7** (a) The nervous system sends electrical impulses to parts of the body in response to an external stimulus.

(i) Name the structure in the nervous system that detects an external stimulus.

**(1)**

(ii) Name the structure that transmits electrical impulses directly to the central nervous system.

**(1)**

(b) Neurones are specialised cells.

(i) Describe what is meant by a specialised cell.

**(2)**

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## 4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

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(ii) A motor neurone is a specialised cell.

Draw a diagram of a motor neurone.

Your diagram should include these labels.

- nucleus
- cell body
- myelin sheath

**(4)**

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(iii) Describe the function of a motor neurone in the nervous system.

**(2)**

## 4. Question 7 · part c · 4 marks

June 2024 Paper 1R

Further practice · Attempt part c only.

(c) Neurotransmitters allow different neurones to communicate with each other.

The distance between two neurones is 100 nm.

Neurotransmitters diffuse across the gap between neurones at a speed of $4.0 \times 10^6$ nm/s.

Calculate the time taken, in milliseconds, for a neurotransmitter to cross the gap between two neurones.

Give your answer in standard form.

**(4)**

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time taken = ..... ms

**(Total for Question 7 = 14 marks)**

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## 4. Question 7 · part c · 4 marks · Working and answer

Working and answer

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## 5. Question 1 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

**1** (a) The sentences are about the heart.

Complete the sentences by using words or numbers from the box.

**(8)**

| | | | | | | |
|---|---|---|---|---|---|---|
| 80 | 100 | 120 | atria | body | diastolic | elastic tissue |
| lungs | muscle | systolic | tendon | vena cava | ventricles | |

The heart consists of four chambers.

The two upper chambers are called ..... .

The left upper chamber receives blood from the ..... whilst the right upper chamber receives blood from the body through the ..... .

The two lower chambers are called ..... .

The left lower chamber pumps blood to the ..... .

This left lower chamber has a very thick wall made of ..... .

When the chamber contracts, it generates a pressure known as the ..... pressure.

The average value for this pressure is about ..... mmHg.

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## 5. Question 1 · part b · 3 marks

June 2025 Paper 1R

Further practice · Attempt part b only.

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(b) Cardiac output is the volume of blood pushed out of the heart in one minute.

Cardiac output is calculated using this formula.

cardiac output = heart rate × stroke volume

The stroke volume is the volume, in $dm^3$, of blood pushed out during one heartbeat. The heart rate is the number of beats per minute.

Calculate the stroke volume for a person with a heart rate of 68 beats per minute and a cardiac output of 6.1 $dm^3$.
Give your answer in $cm^3$.

**(3)**

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stroke volume = ..... $cm^3$

**(Total for Question 1 = 11 marks)**

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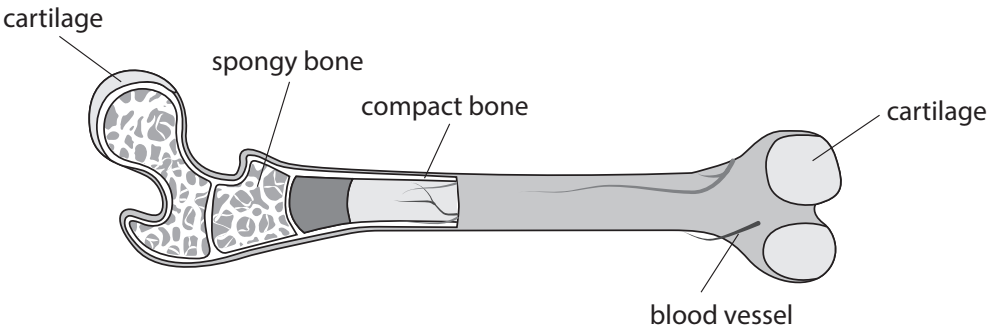
Working and answer

# 6. Question 1 · part b.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

**1** (a) The diagram shows a bone found in the human body.




(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

**(1)**

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

**(3)**

| Part | Function | | |
|---|---|---|---|
| | reduces friction | where red blood cells are produced | provides strength |
| compact bone | | | |
| spongy bone | | | |
| cartilage | | | |

(iii) Describe the function of the blood vessel in the bone.

**(2)**

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# 6. Question 1 · part b.ii · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.ii only.

(b) Bones are moved by muscles.

(i) Use the words in the box to complete the passage about bones and muscles.

**(4)**

| | | | | |
|---|---|---|---|---|
| cartilage | contracts | ligaments | lowers | pulls |
| pushes | raises | relaxes | tendons | |

Muscles are attached to bones by ..... .

To raise the lower arm, the biceps ..... and the triceps ..... .

The biceps ..... on the radius, which raises the lower arm.

(ii) Energy is transferred by the biceps muscle when raising the lower arm.

The biceps muscle applies a force of 125 N to raise the lower arm a distance of 0.50 m.

Use this formula to calculate the energy transferred by the biceps muscle when raising the lower arm 0.50 m.

energy transferred (J) = force (N) × distance (m)

**(2)**

energy transferred = ..... J

**(Total for Question 1 = 12 marks)**

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## 6. Question 1 · part b.ii · 2 marks · Working and answer

Working and answer

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## 7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

**3** (a) The nervous system is made up of sensory, motor and relay neurones.

(i) Draw a diagram of a motor neurone.

You should label these structures in your diagram.

- nucleus
- axon
- myelin sheath

**(4)**

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(ii) Describe the function of a motor neurone.

**(2)**

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# 7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

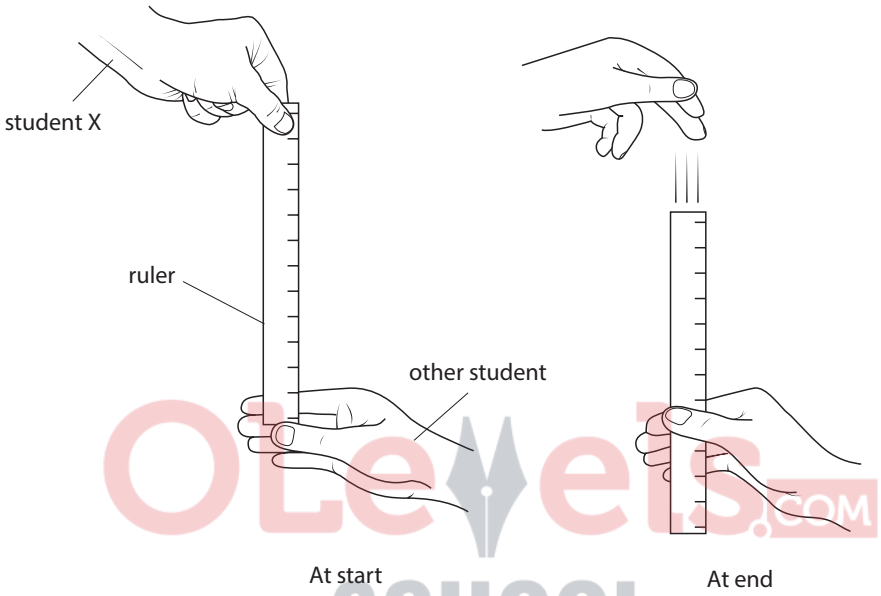
Further practice · Attempt part b.iv only.

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(b) Student X uses this equipment to investigate the reaction time of another student.




Student X knows that the reaction time can be calculated using this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

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# 7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

(i) Describe a method the student could use to compare the reaction times of 10 students.

**(6)**

(ii) Name the dependent variable in this investigation.

**(1)**

(iii) Give a reason why the student should display the results using a bar chart instead of a line graph.

**(1)**

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# 7. Question 3 · part b.iv · 2 marks

June 2025 Paper 2R

Further practice · Attempt part b.iv only.

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(iv) The reading on the ruler for one of the students is 0.15 m.

Calculate the reaction time of this student.

Use this formula.

$$\text{reaction time} = \sqrt{\frac{\text{measurement from ruler}}{5}}$$

**(2)**

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reaction time = ..... s

**(Total for Question 3 = 16 marks)**

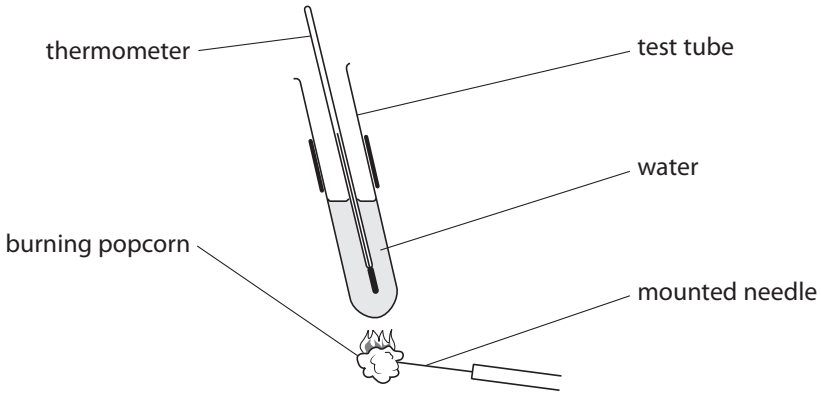
Working and answer

# 8. Question 2 · part b.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

**2** A student uses this apparatus to investigate the energy contained in popcorn.




This is the student's method.

- place 12 g of water in a test tube
- measure the initial temperature of the water
- place the popcorn on a mounted needle
- light the popcorn using a Bunsen flame
- place the burning popcorn under the test tube
- measure the maximum temperature of the water

(a) (i) State why the test tube should have thin glass walls.

**(1)**

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(ii) Explain one safety precaution that the student should take during the investigation.

**(2)**

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# 8. Question 2 · part b.i · 3 marks

November 2020 Paper 1

Further practice · Attempt part b.i only.

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(b) The student uses this equation to calculate the energy released by the popcorn.

energy released = mass (g) × 4.2 × temperature rise (°C)

The initial temperature of the 12 g of water is 17 °C.

The maximum temperature of the water is 45 °C.

(i) Calculate the energy released when the student burns the popcorn.

Give your answer to 2 significant figures.

**(3)**

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energy released = ..... J

(ii) Explain why the calculated amount of energy released is not the same as the full amount of energy contained in the popcorn.

**(3)**

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## 8. Question 2 · part b.i · 3 marks · Working and answer

Working and answer

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# 9. Question 7 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

**7** The volume of alcohol in alcoholic drinks can be measured in units.

One unit is equivalent to 10 $cm^3$ of pure alcohol.

The diagram shows a bottle of wine with 14% alcohol content and a glass containing 250 $cm^3$ of this wine.




(a) Calculate the number of units of alcohol in the glass of wine.

**(2)**

number of units = .....

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# 9. Question 7 · part b.i · 2 marks

June 2019 Paper 2

Further practice · Attempt part b.i only.

(b) It is illegal for a person to drive a motor vehicle if they have more than the legal limit of alcohol in their bloodstream.

It takes one hour for the body to process one unit of alcohol so that there is no alcohol left in the bloodstream.

(i) A person drinks three units of alcohol. This puts her at the legal limit.

Calculate the time it would take before a person can legally drive after drinking four glasses of Ruby Red wine.

**(2)**

time = ..... hours

(ii) Suggest why it is illegal for people to drive a motor vehicle with more than the legal limit of alcohol in their bloodstream.

**(2)**

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## 9. Question 7 · part b.i · 2 marks · Working and answer

Working and answer

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